1. Problem Definition

This project builds a machine learning model to predict whether a person earns more than $50,000 per year based on demographic and employment data from the UCI Adult Census dataset. The business goal is to maximize precision - minimize wasted outreach by only contacting people who are very likely high earners. Each outreach contact costs money, and contacting low earners wastes the budget without generating returns.

The dataset contains 48,842 records with 14 raw features (age, workclass, education, capital gain, hours worked, etc.) and a binary target variable (income >50K or <=50K). The positive class (~24%) is significantly underrepresented, making precision a more informative metric than accuracy.

2. Data Preparation

Data was loaded from the UCI Machine Learning Repository. Missing values (marked as ' ?') were dropped. The dataset was split into an 80% training set and 20% test set using stratified sampling (random_state=42) to preserve the 76/24 class balance. Within the training set, 5-fold Stratified cross-validation was used for all model evaluation and hyperparameter search.

Seven engineered features were created: log_capital_gain, has_capital_gain, has_capital_loss, age_group, hours_category, higher_ed, and edu_hours_interaction. These captured patterns raw data alone could not express. SelectKBest(mutual_info_classif, k=30) selected the 30 most informative features from ~122 one-hot encoded columns.

3. Model Development

Three models were compared with identical preprocessing pipelines: Logistic Regression, Random Forest, and

HistGradientBoostingClassifier. Cross-validation precision scores were: Logistic Regression 0.7666, Random Forest 0.8018, and HistGradientBoosting 0.8025 - narrowly winning.

HistGradientBoosting was selected because it consistently outperformed across all metrics and showed the lowest variance across folds. Its sequential tree-building approach, where each tree corrects previous mistakes, proved more effective than Random Forest's parallel voting for this imbalanced classification problem.

4. Model Tuning

Hyperparameter tuning used RandomizedSearchCV with 5-fold Stratified CV, optimizing for precision. Budget: 50 iterations for LR and HGB, 30 for RF, totaling ~30 minutes. Best HGB hyperparameters: learning_rate=0.01432, max_iter=127, max_depth=10, l2_regularization=0.0152, min_samples_leaf=16.

5. Diagnostics

Learning curve analysis confirmed a GOOD FIT: training precision 0.8401 (+/- 0.0038), validation precision 0.8012 (+/- 0.0082), gap 0.0389. Validation curves confirmed diminishing returns beyond selected settings. Isotonic calibration (5-fold CV) improved Brier score from 0.0949 to 0.0918, making probability statements trustworthy.

6. Final Results

Final evaluation on untouched 20% test set (6,513 records) at threshold 0.832:

Precision: 0.9695 | Recall: 0.3036 | F1: 0.4624 | Specificity: 0.997 | Brier: 0.0918 | ROC AUC: 0.9213

Confusion matrix: TP=476, FP=15, FN=1092, TN=4930. The model flagged 491 people as high earners, only 15 false alarms - 96.95% precision. We miss 1,092 true high earners but waste money on only 15 contacts instead of hundreds. For a precision-focused business, this is the preferred operating point.

7. Model Interpretation

Permutation importance ranked top features: marital_status (0.0524), capital_gain (0.0502), education_num (0.0352), age (0.0222), capital_loss (0.0132). Engineered features contribute through parent original features. fnlwgt had NEGATIVE importance (-0.0072) - the model correctly avoided noise. workclass, relationship, native_country had near-zero importance. Domain knowledge-driven features consistently outperformed raw features.

8. Production Readiness

The complete pipeline saved as day4-final-pipeline.joblib (engineer -> preprocessor -> select -> model). Deployment requires loading ONE file and calling .predict_proba() on raw data - no manual preprocessing. Optimal threshold 0.832 stored in day4-threshold-info.json. Reproducibility ensured by fixed random_state=42 everywhere, documented library versions in requirements.txt, and step-by-step README.

9. Limitations & Future Improvements

Limitations: (1) Recall only 0.3036 - ~70% of true high earners missed; (2) Saved pipeline is raw model while threshold derived from calibrated probabilities - minor inconsistency; (3) Only UCI Adult dataset tested.

Future improvements: (1) More training data to improve recall; (2) Subgroup-specific thresholds; (3) Remove noise features like fnlwgt; (4) Test on different regions; (5) SHAP values for per-prediction explainability; (6) API wrapper for real-time deployment.

Week 1 Workflow Summary

Day 1: Built baseline models with fully reproducible pipelines. Fixed random states, documented library versions, verified pipeline structure.

Day 2: Performed hyperparameter search using RandomizedSearchCV across all three models. HGB won with CV precision 0.8025.

Day 3: Created 7 engineered features, rebuilt pipeline with FunctionTransformer. Confirmed HGB wins CV comparison (0.7758 +/- 0.0073). Applied SelectKBest(k=30).

Day 4: Isotonic calibration improved Brier 0.0949 to 0.0918. Found optimal threshold 0.832. Achieved final test precision 0.9695.

Day 5: Validated final model, performed error analysis, extracted feature importance, built inference workflow, and documented the complete project.

Week 1 Key Results

Model / Phase	Precision	Key Detail
Baseline HGB	0.7758	CV, Day 3 baseline
Tuned HGB	0.8025	CV, best hyperparameters
Calibrated HGB	0.8025	Brier: 0.0918
Final Test	0.9695	Untouched test set
Threshold	0.832	Max precision, recall>=0.30
Top Features	---	marital_status, capital_gain